

Optimizing HPC Performance for CDC-Based Storage and Reconstruction of High-Resolution Imagery

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Abstract—High-resolution imagery is used more often for community asset management, environmental monitoring, and disaster response. These images preserve fine details such as buildings, roads, vehicles, vegetation, and damage patterns, but they also create a storage problem. A single field mission can generate hundreds of gigabytes of data, which increases long-term storage and transfer costs. To reduce this storage burden, this study evaluates an existing conditional diffusion compression (CDC) workflow for high-resolution image storage and reconstruction on GPU supercomputers. We do not propose a new compression model. Instead, we study how CDC behaves as an HPC workload and identify practical operating points that balance compression ratio, GPU memory use, reconstruction speed, image quality, and downstream analysis utility. Using 100 RGB drone images from Galveston, Texas, we test different CDC checkpoints and tiled reconstruction settings on NCSA DeltaAI GH200. The balanced checkpoint with 256×256 tiled reconstruction reduces peak GPU memory from 51.96 GB to 1.57 GB and lowers reconstruction time from 143.17 to 79.34 seconds per image compared with full-image reconstruction. This setting keeps a $79.98\times$ compression ratio while preserving useful image quality and downstream detection and segmentation performance. The results show that CDC can support long-term storage of high-resolution imagery when reconstruction is treated as a system optimization problem. A practical HPC setting should report on storage savings, GPU memory, throughput, image quality, visible artifacts, and downstream utility together.

I. SUMMARY

Drone imagery is valuable after disasters because it preserves roof damage, vehicle access, road conditions, and other fine visual details. Those same details make the data hard to use at scale. A single collection can create hundreds of gigabytes of imagery, and repeated reconstruction can turn a storage problem into a GPU scheduling problem. For SC, the core question is therefore a systems question: how should a data-centric AI workflow choose a compression and reconstruction operating point that fits memory, queue time, storage constraints, and downstream analysis needs?

We treat CDC as an existing reconstruction path, not as an original algorithmic contribution. The contribution is a measured HPC study of how checkpoint choice and tiled reconstruction change resource behavior on NCSA DeltaAI GH200. The workflow processes 100 RGB drone images

TABLE I
SELECTED FULL-RESOLUTION DELTA AI OPERATING POINTS ON 50
DRONE IMAGES.

Setting	Sec./img	Img/hr	GB	Ratio	F1	Roof IoU
High quality 512	88.51	40.7	2.96	$29.73\times$	0.8805	0.9001
Balanced 256	79.34	45.4	1.57	$79.98\times$	0.8705	0.8989
Max comp. 256	78.91	45.6	1.57	$139.89\times$	0.8679	0.8956

from Galveston, Texas. For the main N50 tradeoff matrix, each configuration records wall time, images per hour, peak GPU memory, bits per pixel, compression ratio, PSNR, SSIM, LPIPS, seam-related artifacts, and visual error maps. Object detection and SAM masks are used only as utility checks. They help verify that a resource-efficient setting does not destroy the downstream signals that emergency managers would later inspect.

The central result is that full-image diffusion reconstruction is feasible but poorly matched to routine HPC use. With the balanced checkpoint b00064, full-image reconstruction takes 143.17 seconds per image and peaks at 51.96 GB of GPU memory. The same checkpoint with 256×256 tiled reconstruction takes 79.34 seconds per image, reaches 45.4 images per hour on one GH200 GPU, and peaks at 1.57 GB. This is a 44.6% runtime reduction and more than a $30\times$ reduction in peak GPU memory relative to the full-image balanced run. It also keeps a $79.98\times$ compression ratio, 33.14 dB PSNR, 0.8768 SSIM, and 0.001826 LPIPS. The 512×512 tile remains a quality-safe backup, but it is slower and uses more memory than the 256×256 setting.

The downstream measurements support this operating-point choice without turning the poster into a detection paper. On human vehicle and roof labels, original images reach 0.8829 F1. Balanced 256×256 reconstruction reaches 0.8705 F1, a relative drop of about 1.4%, while maximum compression reaches 0.8679 F1. The stricter mAP@0.5:0.95 metric is more sensitive, which means localization quality should be reported separately from classification stability. SAM is used similarly as a mask-stability check. All 1,013 vehicle prompts and 1,201 roof prompts return usable masks. For balanced 256×256 ,

vehicle masks reach 0.7346 mean IoU and 0.8380 Dice, while roof masks reach 0.8989 mean IoU and 0.9447 Dice.

The poster’s claim is thus not that CDC is a new model. The claim is that CDC can be made practical for full-resolution drone imagery on a GPU supercomputer when it is treated as a systems operating-point problem. A useful setting must report storage savings, GPU memory, reconstruction throughput, scalar quality, visible artifacts, object-level utility, and segmentation stability together. Balanced checkpoint b00064 with 256×256 tiling is the recommended setting because it converts a memory-heavy diffusion job into a low-memory, higher-throughput workflow while preserving disaster-relevant image utility.